EXPERIMENT-3

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1. COMMAND LINE INTERFACE:

PYTHON CODE:

```
import psutil
import time
import os
import sys

def clear_screen():
    if os.name == "nt":
        os.system("cls")
    else:
        os.system("clear")

def get_stats():
    cpu = psutil.cpu_percent(interval=1)
    ram = psutil.virtual_memory().percent
    disk = psutil.disk_usage('/').percent

    battery = psutil.sensors_battery()
    if battery:
        status = "Charging" if
battery.power_plugged else "Not Charging"
        battery_text = f"{battery.percent} %
({status})"
    else:
        battery_text = "Not Available"

    return cpu, ram, disk, battery_text

try:
    while True:
        clear_screen()
        cpu, ram, disk, battery = get_stats()

        print("=" * 30)
        print(" SYSTEM PERFORMANCE (CLI)")
        print("=" * 30)
        print(f"CPU Usage     : {cpu} %")
        print(f"RAM Usage     : {ram} %")
        print(f"Disk Usage : {disk} %")
        print(f"Battery Level : {battery}")
        print("\nPress Ctrl+C to exit")

        time.sleep(1)
```

```
except KeyboardInterrupt:
    clear_screen()
    print("CLI System Monitor exited.")
    sys.exit()
```

OUTPUT:

2. GRAPHICAL USER INTERFACE:

PYTHON CODE:

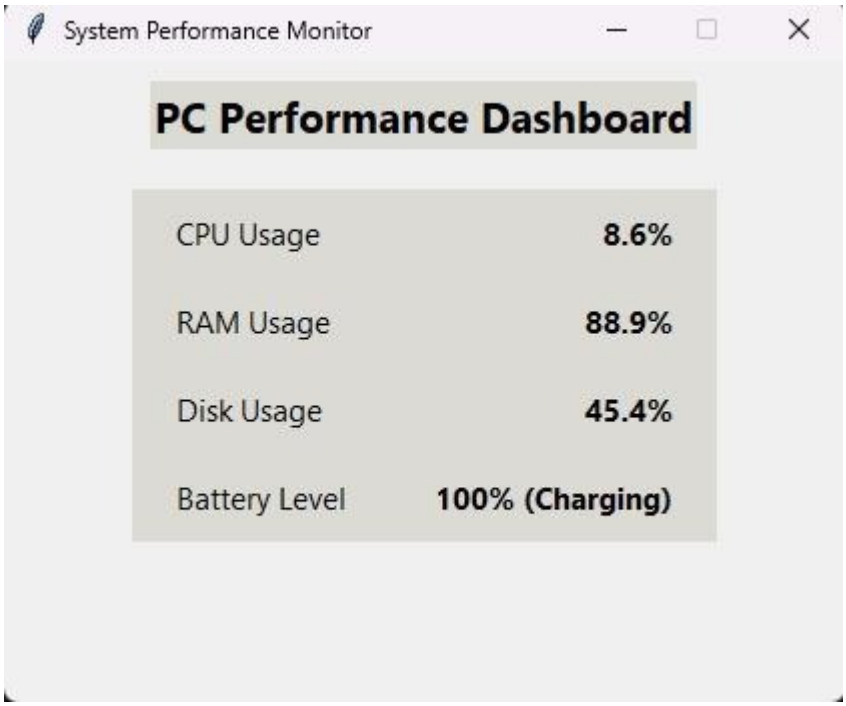
```
import tkinter
as tk
from tkinter
import ttk
import psutil



#-------------------
GUI SETUP ------
-------------
root = tk.Tk()
root.title("Syste
m Performance
Monitor")
root.geometry(
"420x320")
root.resizable(F
alse, False)



style =
ttk.Style()
style.theme_us
e("clam")


#-------------------
FUNCTIONS -----
--------------
def
update_stats():
    # CPU
    cpu =
psutil.cpu_perc
ent()

cpu_label.confi
g(text=f"{cpu}%
```

```
")

    # RAM
    ram = psutil.virtual_memory().percent
    ram_label.config(text=f"{ram}%")

    # Disk
    disk = psutil.disk_usage('/').percent
    disk_label.config(text=f"{disk}%")

    # Battery
    battery = psutil.sensors_battery()
    if battery:
        status = "Charging" if battery.power_plugged else "NotCharging"
        battery_label.config(text=f"{battery.percent}% ({status})")
    else:
        battery_label.config(text="Not Available")
```

```
root.after(1000, update_stats)  # update every second


#-------------------UILAYOUT -------------------------
title = ttk.Label(root, text="PC Performance Dashboard", font=("Segoe UI", 16, "bold"))
title.pack(pady=10)


frame = ttk.Frame(root)
frame.pack(pady=10)


def stat_row(name, row):
    label = ttk.Label(frame, text=name, font=("Segoe UI", 11))

label.grid(row=row, column=0, padx=20, pady=10, sticky="w")


    value =
```

```
ttk.Label(frame,
text="0%",
font=("Segoe
UI", 11, "bold"))


value.grid(row=
row,  column=1,
padx=20,
pady=10,
sticky="e")

    return value



cpu_label =
stat_row("CPU
Usage", 0)

ram_label =
stat_row("RAM
Usage", 1)

disk_label =
stat_row("Disk
Usage", 2)

battery_label =
stat_row("Batte
ryLevel", 3)



#-------------------
START  ------------
-------

update_stats()

root.mainloop()
```

OUTPUT:



3. VOICE USER INTERFACE:

PYTHON CODE:

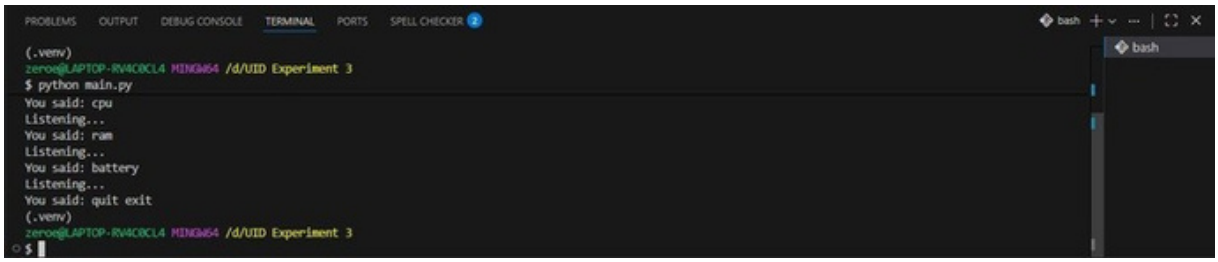
```
import psutil
import pyttsx3
import speech_recognition
    as sr

engine = pyttsx3.init()

def speak(text):
    engine.say(text)
    engine.runAndWait()

def listen():
    r = sr.Recognizer()
    with sr.Microphone() as
    source:
        print("Listening...")

    r.adjust_for_ambient_noi
    se(source)
        audio = r.listen(source)
```

```
    try:
        command =
    r.recognize_google(audio)
    .lower()
        print("You said:",
    command)
        return command
    except:
        return ""

defcpu_usage():
    usage =
    psutil.cpu_percent(interv
    al=1)
    speak(f"CPU usage is
    {usage} percent")

def ram_usage():
    ram =
    psutil.virtual_memory().p
    ercent
    speak(f"RAM usage is
    {ram} percent")

def disk_usage():
    disk =
    psutil.disk_usage('/').perc
    ent
    speak(f"Disk usage is
    {disk} percent")

def battery_status():
    battery =
```

```
    psutil.sensors_battery()
    ifbattery:
    status = "charging" if
    battery.power_plugged
    else "not charging"
    speak(f"Battery level is
    {battery.percent} percent
    and it is {status}")
    else:
    speak("Battery
    information is not
    available")

speak("Voice system
    monitor activated")

whileTrue:
    command = listen()

    if"cpu" in command:
        cpu_usage()

    elifany(w in command for
    win["ram", "memory",
    "mem"]):
        ram_usage()

    elif"disk" in command or
    "storage" in command:
        disk_usage()

    elif"battery" in command
    or"power" in command:
        battery_status()
```

```
    elif "exit" in command or
    "quit" in command:
        speak("Shutting down
    system monitor")
        break
```

OUTPUT :



4. USER SATISFACTION COMPARISON:

PYTHON CODE:

```
def survey():
    print("Rate your satisfaction with the following interfaces (1-5):")

    # Get user input for each interface
    try:
        cli_satisfaction = int(input("CLI (1-5): "))
        gui_satisfaction = int(input("GUI (1-5): "))
        vui_satisfaction = int(input("VUI (1-5): "))

        # Ensure valid ratings
        if not (1 <= cli_satisfaction <= 5 and 1 <= gui_satisfaction <= 5 and 1 <= vui_satisfaction <= 5):
            print("Please enter ratings between 1 and 5 only.")
            return

        # Display the ratings
        print("\nYour satisfaction ratings:")
        print(f"CLI: {cli_satisfaction}")
```

```
print(f"GUI: {gui_satisfaction}")
```

```
        print(f"VUI: {vui_satisfaction}")

        # Calculate the average satisfaction
        avg_satisfaction = (cli_satisfaction + gui_satisfaction + vui_satisfaction) / 3
        print(f"\nAverage Satisfaction Score: {avg_satisfaction:.2f}")

    except ValueError:
        print("Invalid input! Please enter numbers between 1 and 5.")

# Run the survey function
if __name__ == "__main__":
    survey()
```

```
D:\230701304>python survey.py
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 4
GUI (1-5): 5
VUI (1-5): 3

Your satisfaction ratings:
CLI: 4
GUI: 5
VUI: 3

Average Satisfaction Score: 4.00

D:\230701304>
```